Right Triangles in Elevation and Depression Problems Answer Key

High School Geometry

Quick Answers

1. 24.99
2. 399.9
3. 8.5
4. $\frac{8}{15}$
5. 70.42
6. 11943
7. 10.98
8. 65.97

Common wrong answers

1. *If they got 21.2: used sine, which treats the 40 ft ground distance as the hypotenuse*
2. *If they got 18.07: multiplied by the tangent instead of dividing by it*
3. *If they got 81.47: inverted the ratio and used run over rise, 20/3, in the inverse tangent*
4. *If they got 1.875: wrote the ratio upside down, adjacent over opposite (45/24, i.e. 15/8)*
5. *If they got 79.76: used tangent, which treats the 150 ft taut string as a ground distance*
6. *If they got 857.44: multiplied the altitude by the tangent instead of dividing by it*
7. *If they got 9.38: found the height above eye level only and forgot to add the 1.6 m eye height*
8. *If they got 186.6: subtracted the angles first and worked one triangle at 15 degrees*
8. *If they got 208.78: added the two horizontal distances instead of subtracting them*

1. Tree from a ground distance — elevation, solve for a side.

Given: horizontal distance 40 ft (adjacent to the angle), angle of elevation 32°. **Wanted:** the tree height h, which is opposite that angle.

Choose the ratio. Opposite and adjacent are the two sides in play, so tangent is the tool: $\tan 32° = \dfrac{h}{40}$.

Solve. $h = 40 \tan 32° = 40(0.624869\ldots) = 24.9948\ldots$ $\boxed{24.99 \text{ ft}}$

Common wrong answer: 21.20 ft — from $40 \sin 32°$, which treats the 40 ft ground distance as the hypotenuse. On level ground the hypotenuse is the slanted line of sight, never the ground.

2. Boat from a lighthouse — depression, solve for a side.

Given: height 85 m, angle of depression 12°. **Wanted:** the horizontal distance d from the foot of the lighthouse.

Move the angle into the triangle. The angle of depression is measured from the horizontal at the top. That horizontal is parallel to the sea, so the angle of elevation from the boat to the lamp equals it: 12° (alternate interior angles). Inside the right triangle, 85 m is opposite 12° and d is adjacent.

Choose the ratio and solve. $\tan 12^\circ = \dfrac{85}{d} \Rightarrow d = \dfrac{85}{\tan 12^\circ} = \dfrac{85}{0.212556\ldots} = 399.8936\ldots$

$\boxed{399.9 \text{ m}}$

Common wrong answer: 18.07 m — multiplying by the tangent instead of dividing. A sanity check kills it: a 12° sight line is very shallow, so the boat must be far out, not a fifth of the lighthouse's height away.

3. Ramp angle — elevation, solve for the angle.

Given: rise 3 ft (opposite), run 20 ft (adjacent). **Wanted:** the angle θ itself.

Choose the ratio. Two sides known, angle wanted — so build the ratio, then undo it with the inverse function: $\tan\theta = \dfrac{3}{20} = 0.15$.

Solve. $\theta = \tan^{-1}(0.15) = 8.5308\ldots^\circ$ $\boxed{8.5^\circ}$

Common wrong answer: 81.47° — from $\tan^{-1}(20/3)$, the ratio built upside down. That is the angle the ramp makes with the *vertical*; a ramp you could not walk up is the tell.

Note for grading: $\tan^{-1}$ is the inverse tangent, not $1/\tan$. A student who computes $1/\tan(0.15)$ gets nothing resembling an angle. (Real ramps are built at about 4.8°, so this one is steep.)

4. Write the ratio — no solving.

Given: the top of the tower is 24 m above eye level (opposite the angle of elevation) and 45 m away horizontally (adjacent).

Set up. By definition, $\tan\theta = \dfrac{\text{opposite}}{\text{adjacent}} = \dfrac{24}{45}$.

Reduce. Both are divisible by 3: $\boxed{\frac{8}{15}}$

Common wrong answer: the reciprocal, $\dfrac{45}{24} = \dfrac{15}{8} = 1.875$. That is $\cot\theta$ — adjacent over opposite — and it describes the same triangle from the other angle.

Why this matters: the ratio is the exact answer. Turning it into $\theta \approx 28.07^\circ$ throws away exactness and was not asked for.

5. Kite on a taut string — elevation, hypotenuse given.

Given: string length 150 ft — the *hypotenuse*, because the string is the line of sight — and an angle of elevation of 28°. **Wanted:** the height, opposite that angle.

Choose the ratio. Opposite with hypotenuse means sine, not tangent: $\sin 28° = \dfrac{h}{150}$.

Solve. $h = 150 \sin 28° = 150(0.469472\ldots) = 70.4207\ldots$ $\boxed{70.42 \text{ ft}}$

Common wrong answer: 79.76 ft — from $150 \tan 28°$, which treats the taut string as a ground distance. The height of a kite can never exceed the length of its string, and tangent will happily break that rule.

Teaching note: problems 1 and 5 look identical in words. The difference is which side the given length is — ground distance (adjacent, use tangent) versus taut string (hypotenuse, use sine).

6. Plane and airport — depression, solve for a side.

Given: altitude 3200 ft, angle of depression 15°. **Wanted:** ground distance d from the point directly beneath the plane.

Move the angle into the triangle. As in problem 2, the angle of elevation from the airport equals the angle of depression from the plane: 15°. The altitude is opposite it; d is adjacent.

Solve. $d = \dfrac{3200}{\tan 15°} = \dfrac{3200}{0.267949\ldots} = 11942.56\ldots$ $\boxed{11943 \text{ ft}}$

Common wrong answer: 857 ft — multiplying instead of dividing. That would put the airport closer than a quarter mile while the plane is still 3200 ft up, which no 15° glide can do.

7. Flagpole measured from eye level — elevation with an offset.

Given: eye height 1.6 m, horizontal distance 12 m, angle of elevation 38°. **Wanted:** the height of the pole *above the ground*.

Locate the triangle. The angle is measured at the eye, so the right triangle sits entirely *above eye level*: its horizontal leg is 12 m and its vertical leg is the part of the pole above eye level, call it y.

Solve the triangle. $y = 12 \tan 38° = 12(0.781286\ldots) = 9.3754\ldots$ m.

Return to the question. The pole also runs from the ground up to eye level, so add that back: $1.6 + 9.3754\ldots = 10.9754\ldots$ $\boxed{10.98 \text{ m}}$

Common wrong answer: 9.38 m — stopping at the triangle instead of answering the question asked. The 1.6 m is the one number on this sheet that does not belong inside a trig ratio.

8. Two boats from one cliff — two triangles, shared leg.

Given: cliff height 50 m; angles of depression 35° (nearer boat) and 20° (farther boat).
Wanted: the distance between the boats.
Set up two triangles. Both are right triangles with the same vertical leg of 50 m, and both angles transfer to the boats as angles of elevation. Let d_{20} and d_{35} be the horizontal distances from the foot of the cliff.
Solve each.

$$d_{20} = \frac{50}{\tan 20^\circ} = \frac{50}{0.363970\ldots} = 137.3739\ldots \text{ m}$$
$$d_{35} = \frac{50}{\tan 35^\circ} = \frac{50}{0.700207\ldots} = 71.4074\ldots \text{ m}$$

Combine. The boats are on the same side and in line with the cliff, so the gap is the difference: $137.3739\ldots - 71.4074\ldots = 65.9665\ldots$ $\boxed{65.97 \text{ m}}$

Common wrong answer: 186.60 m — subtracting the angles first ($35^\circ - 20^\circ = 15^\circ$) and working a single triangle. Angles cannot be subtracted that way: the two sight lines start at the same point but land on different triangles, so the *distances* are what get subtracted.

Common wrong answer: 208.78 m — adding the two distances. That is the answer to a different picture, with the boats on opposite sides of the cliff; here they are both out to sea in the same direction.

Check the smaller angle gives the larger distance: a shallower look reaches further out, and $137.37 > 71.41$ confirms it.